Investigator's Brochure (Abridged)

Xanomeline-Trospium Combination Therapy

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1. Introduction

Xanomeline is a potent muscarinic acetylcholine receptor agonist with high affinity for M1 and M4 receptor subtypes. Unlike cholinesterase inhibitors, which indirectly increase acetylcholine levels, xanomeline directly stimulates muscarinic receptors. Preclinical studies have shown that M1 receptor activation enhances cognitive function through modulation of hippocampal and cortical neural circuits. M4 receptor activation may reduce psychotic symptoms common in Alzheimer's disease.

2. Pharmacokinetics

Following oral administration, xanomeline reaches peak plasma concentrations in 1-2 hours. The elimination half-life is approximately 4-6 hours. Steady-state concentrations are achieved within 3-5 days of twice-daily dosing. Xanomeline is extensively metabolized by CYP2D6 and CYP3A4 with less than 1% excreted unchanged in urine. Trospium, when co-administered, does not significantly affect xanomeline pharmacokinetics. Food intake reduces the rate but not the extent of xanomeline absorption.

3. Preclinical Safety

In preclinical toxicology studies, the primary findings were related to the peripheral cholinergic effects of xanomeline. In rats, doses up to 30 mg/kg/day for 6 months showed evidence of gastrointestinal irritation at the highest dose. No carcinogenic potential was identified in 2-year bioassays. Reproductive toxicity studies showed no evidence of teratogenicity at clinically relevant exposures. Cardiovascular safety pharmacology studies demonstrated no effects on hERG channel current or cardiac action potential duration at concentrations up to 10 microM.